FIGURE 3: CARD CATALOG USER STORIES

## PANEL A: USER STORY 1 — NEW HIRE ONBOARDING WORKFLOW
Jordan Reyes: Getting Oriented on ROS

GOAL: First week at a new lab — ROS comes up constantly. Figure out, in plain terms, what it actually is, and how it fits the field's other major post-mortem brain-bank studies.

PHASE 1: CRYSTALLIZE — WHAT IS ROS?
HOW: Publications -> filter Resource Name = Religious Orders Study -> Freeze -> AI Analysis

Publications Page (4,979 papers) ——— > Freeze (205 rows) -> 205 papers — 58% also tag Memory and Aging. Top co-resource: MAP.

> AI Analysis -> ROS and MAP are near-identical sister cohorts — routinely analyzed together as ROSMAP.

PHASE 2: STRATIFY BY GENE (AI Analysis's next step B)
HOW: Connections -> +Join SciLite Annotations on the 205 ROS papers (facet: Type=Gene_Proteins) -> count per gene (Exact)

ROS papers (205, from Phase 1) ——— > Join by PMC ID (Gene facet) -> 157/205 papers tag a gene: tau 105 · APOE 95 · APP 48 · TDP-43 30 · GFAP 28 · TREM2 26

OUTCOME: ROS's gene/protein signal is the canonical AD molecular axis — tau and APOE lead, with APP, TDP-43, GFAP and TREM2 close behind. The new hire now knows ROS reads as post-mortem neuropathology evidence, not a living-cohort biomarker signal.

## PANEL B: USER STORY 2 — BIOMEDICAL RESEARCHER WORKFLOW
Dr. Sarah Chen: From Hypothesis to Publication

RESEARCH QUESTION: Role of microglial dysfunction in early-stage Alzheimer's?

PHASE 1: DISCOVERY — IS THERE ACTUALLY A GAP?
HOW: Connections -> +Join SciLite Annotations (facet: Type=Gene_Proteins, Exact=TREM2) onto Publications -> +Join Resources (join key: Resource Name, facet: Diseases Included = Alzheimer's Disease or Preclinical Alzheimer's Disease) -> Freeze -> read the Scientific Read's own Contrastive section

215 AD/Preclinical-linked TREM2 papers — dominated by postmortem-tissue and genomics infrastructure (AMP-AD, ROS/MAP family, ADGC, ADSP). Swedish BioFINDER-2 — the catalog's 2nd-largest resource overall — doesn't even crack the top 15 here.

PHASE 2: IS THIS TREM2-SPECIFIC, OR IS BIOFINDER RARE FOR EVERY GENE?
HOW: repeat the Phase 1 join for APOE -> compare

Same query, same disease restriction, 1,221 APOE papers: BioFINDER-2 IS the #1 resource here, at 9% of the subset. Same catalog, same cohort — APOE uses it constantly, TREM2 barely at all (3 papers).

PHASE 3: LOOKING FOR A METHOD, NOT JUST A GAP
HOW: same TREM2/AD papers -> +Join Software (join key: Publication) -> read the AI Read's "Notable Outliers"

2 papers (same tissue bank, overlapping author team) use MIBI-TOF spatial proteomics (Angelo lab's open-source toffy + ark-analysis) to image TREM2 protein directly in postmortem microglia — seeing the receptor a GWAS/transcriptomics literature can only infer.

OUTCOME (Dr. Chen): TREM2 showed the gap; these two papers showed a way to close it. My proposal isn't "build new infrastructure" — it's "borrow this pipeline, point it two years earlier."

## PANEL C: USER STORY 3 — PROGRAM OFFICER WORKFLOW
Dr. Michael Torres: Portfolio Analysis & Strategic Planning

GOAL: A peer philanthropic funder just entered this space — a research direction worth tracking, or a sign we're behind?

PHASE 1: WHAT IS GATES VENTURES ACTUALLY INTERESTED IN?
HOW: Connections -> +Join Grants (facet: funder_name = Gates Ventures) -> +Join Resources (facet: Diseases Included = Alzheimer's Disease or Preclinical Alzheimer's Disease) -> Freeze -> AI Read

26 papers (2024–2026), none with an NIH-style grant number — a private philanthropic initiative, not a federal program. The Preclinical AD tag appears 7x the catalog baseline (27% vs. 3.8%); transcriptomics is LOWER than baseline (27% vs. 38%) despite genetics and proteomics both being elevated — a specific tilt toward fluid and digital biomarkers in presymptomatic populations, not the postmortem-tissue transcriptomics that dominates the field.

OUTCOME (Torres): Gates Ventures is interested in presymptomatic screening while the field's own baseline still leans postmortem-tissue transcriptomics — the same blind spot a researcher independently found from the science side (Panel B).